VECTOR

MICROSAR Ethernet Transceiver Driver

Technical Reference

Marvell 88Q2112

Version 2.0.0

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Status	Released

Document Information

History

Author	Date	Version	Remarks
David Röder	2017-07-10	1.00.00	Creation of document
Mark Harsch	2018-09-03	2.00.00	▶ 88Q2112STORYC-6361: Perform maintenance and reviews to achieve proc 3.0 QM release: TechRef
Martin Heiker	2018-09-27		▶ Add section with limitations of Cable Diagnostics

Table 1-1 History of the document

Reference Documents

No.	Title	Version
[1]	TechnicalReference_EthTrcv_Core	see delivery
[2]	Marvell 88Q2110/88Q2112 Datasheet Doc. No. MV-S110850-00	Rev. A, March 23, 2017, Draft

Table 1-2 Reference documents

Scope of the Document

This complement to the technical reference [1] describes the derivative dependent aspects of the Ethernet Transceiver Driver basis software. Please refer to your Release Notes to get a detailed description of the platform (host, compiler, and transceiver) your Vector Ethernet Bundle has been configured for.



Please note
We have configured the programs in accordance with your specifications in the questionnaire. Whereas the programs do support other configurations than the one specified in your questionnaire, Vector´s release of the programs delivered to your company is expressly restricted to the configuration you have specified in the questionnaire.



Note

This document describes aspects that apply for Ethernet Transceiver Driver modules provided by Vector, which are only used for Marvell 88Q2112 transceivers. For this reason, the corresponding infixes of APIs, macros, parameters etc. are replaced by 88Q2112.

The derivative independent aspects of the Ethernet Transceiver Driver are discussed in [1], where the above-mentioned infix is replaced by the markers <Derivative> and <DERIVATIVE>.

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1 Component History

Component Version	New Features
01.00.xx	Initial version of component
02.00.xx	QM release of component

Table 1-1 Component history

2 Introduction

This document describes the functionality, API and configuration of the Ethernet Transceiver Driver.

Supported AUTOSAR Release*:	4	
Supported Configuration Variants:	pre-compile	
Vendor ID:	ETHTRCV_30_88Q2112_VENDOR_ID	30 decimal (=Vector-Informatik, according to HIS)
Module ID:	ETHTRCV_30_88Q2112_MODULE_ID	73 decimal

* For the precise AUTOSAR Release 4.x please see the release specific documentation.

The Ethernet Transceiver Driver provides hardware independent access to control connected transceivers in a generic way. It offers the functionality to control the mode of operation of connected transceivers as well as to determine their current state, e.g. if events like link status change or bus errors happened.

The transceiver itself is a hardware device, which mainly transforms the logical I/0 signals of the Ethernet Controller to the bus compliant electrical levels, currents and timings.

The transceiver driver considered in this technical reference is designed to operate with Marvell 88Q2110 and 88Q2112 transceivers. This documentation however only covers the special aspects of the driver software covering this kind of transceivers, while all general aspects are covered by the complementary document [1].

3 Hardware Overview

Table 3-1 summarizes the supported derivatives and settings this Ethernet Transceiver Driver can be used on. It also provides information on the documentation used during development of the driver for detailed reference to hardware specific information.

Derivatives	▶ 88Q2110 ▶ 88Q2112
Compiler	Compiler independent
Hardware Manuals	▶ 88Q2110/88Q2112 Datasheet Automotive 100/1000BASE-T1 Transceiver Doc. No. MV-S110850-00 Rev. A March 23, 2017, Draft
Safety Manuals	-
Errata Sheets	-
Specifics	-
Used Registers	See EthTrcv_30_88Q2112_LL_Regs.h
Hardware features related to independence or partitioning	-
Access Mechanism	▶ SMI management interface by utilizing Eth or EthSwt ▶ Ethernet switch internal registers by utilizing EthSwt
Hardware Diagnostics	▶ 6 PHY Test modes ▶ Internal/Remote Loopback mode ▶ Cable diagnostics ▶ Signal quality measurement

Table 3-1 Hardware Overview

4 Derivative Specific Functional Description

4.1 Overview

The general functional description of the transceiver driver is part of the general technical reference [1]. This section gives an overview over which of the features are supported by the specific transceiver driver considered in this document and to what extent. The features marked with ■ in Table 4-1 are available for the Ethernet transceiver driver for 88Q2112 transceivers. If a feature is supported with limitations, it is marked with ■. The limitations are explained in section 4.2.

Supported AUTOSAR Standard Conform Features	
Initialization of the Ethernet Transceiver	■
Setting and getting the Transceiver Mode	■
Transceiver wakeup	■
Restart auto negotiation	■
Get link state	■
Get baud rate	■
Get duplex mode	■
Set test mode	■
Set loopback mode	■
Get signal quality	■
Set TX mode	■
Get cable diagnostics result	■
Get PHY identifier	■
Retrieve version info	■
Report development errors to DET	■
Report production errors to DEM	■

Table 4-1 Supported AUTOSAR Standard Conform Features

4.2 Limitations to supported Features and Functions

4.2.1 PHY Test Modes

The PHY Test modes are explained in detail in the general technical reference [1]. In Table 4-2 all Test modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■. The ones that are not supported respectively with ■.

TestMode value	
ETHTRCV_PHYTESTMODE_NONE	■
ETHTRCV_PHYTESTMODE_1	■
ETHTRCV_PHYTESTMODE_2	■
ETHTRCV_PHYTESTMODE_3	■

ETHTRCV_PHYTESTMODE_4	■
ETHTRCV_PHYTESTMODE_5	■
ETHTRCV_PHYTESTMODE_6	■
ETHTRCV_PHYTESTMODE_7	■

Table 4-2 Supported PHY Test Modes

*) The additional test modes 6 and 7 are only supported if the transceiver is running in 1000BASE-T1 mode.

4.2.2 PHY Loopback Modes

The PHY Loopback modes are explained in detail in the general technical reference [1]. In Table 4-3 all Loopback modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■. The ones that are not supported respectively with ■.

LoopbackMode value	
ETHTRCV_PHYLOOPBACK_NONE	■
ETHTRCV_PHYLOOPBACK_INTERNAL	■
ETHTRCV_PHYLOOPBACK_EXTERNAL	■
ETHTRCV_PHYLOOPBACK_REMOTE	■

Table 4-3 Supported PHY Loopback Modes

4.2.3 PHY TX Modes

The TX modes are explained in detail in the general technical reference [1]. In Table 4-4 all TX modes which are supported by the Ethernet transceiver driver considered in this document are marked with ■. The ones that are not supported respectively with ■.

TxMode value	
ETHTRCV_PHYTXMODE_NORMAL	■
ETHTRCV_PHYTXMODE_TX_OFF	■
ETHTRCV_PHYTXMODE_SCRAMBLER_OFF	■

Table 4-4 Supported PHY Tx Modes

4.2.4 PHY Cable Diagnostics

The Cable Diagnostics are explained in detail in the general technical reference [1].

The 88Q2112 transceiver provides valid Cable Diagnostic results in master mode without any additional set-up. In slave mode the 88Q2112 transceiver should be detached from the master interface to provide valid Cable Diagnostic result. A short can only be detected between the positive and negative MDI signal wires. Detection of a short to ground or power supply is not supported by the 88Q2112 transceiver.

5 Abbreviations

Abbreviation	Description
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
BSW	Basis Software
DEM	Diagnostic Event Manager
DET	Development Error Tracer
Eth	Ethernet Controller Driver
EthIf	Ethernet Interface
EthSwt	Ethernet Switch Driver
EthTrcv	Ethernet Transceiver Driver
HIS	Hersteller Initiative Software
ICU	Input Capture Unit
MICROSAR	Microcontroller Open System Architecture (the Vector AUTOSAR solution)
MII	Media Independent Interface
SMI	Serial Management Interface
Platform	Hardware including host and communication controller (might also be integrated in host) on which the communication stack is implemented.
SWS	Software Specification

Table 5-1 Abbreviations

6 Contact

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